Database (SOC-3020), Fall 2026



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Homework Assignment 2

Issued Monday 12 October 2026. Due as **zipped archive** uploaded into eCLASS by **Tuesday 20 October 2026 at noon** (12:00, system's clock). Hardcopy: none (upon request only for specific cases).

Please note the following:

1. Deadline is *strict*. **Electronic deadline passed without submission implies mark 0 (zero)**. (Equivalently, you will not be admitted to enter the classroom for Quiz 2.)

2. The number of students is: 1 to 3. See next.

2.1 Teams formed by **4 members or more are not permitted** (all students are not admitted to Quiz 2).

2.2 Although teams can be formed by students from any of the sections 003 to 004, it is preferable if students in a team belong to the same section.

3. The homework must be written using LaTeX, either by installing one of your favourite versions[1], or by using OVERLEAF, an easy-to-use online editor for LaTeX available at https://www.overleaf.com.

3.1 Any LaTeX template (OVERLEAF provides many) is okay. Teams are encouraged to create their own template, provided that the cover of their work contains the data required (see next).

3.2 The **cover** must contain the full name of the students in the team, their ID number and section. In particular, students shall be listed in increasing order by ID number; the name of the student who eventually uploads the homework to eCLASS shall be underlined.

4. MODALITY OF SUBMISSION:

+ One member of the team (say, the "team leader") is required to upload into eCLASS one single **zipped archive** containing:
(1) the LaTeX source file (extension .tex);
(2) its compilation (.pdf file),
(3) a folder/directory with all images or figures used in the document (if any),
(4) a file 00X-ID-DDL.sql (see Exercise 3),
(5) a file 00X-ID-DML.sql (see Exercise 3).
+ The **name of zipped archive**, as well the **name of each submitted file** (1) to (5) must be in format: 00X-ID (Example: 001-U2310123), where: 00X is the section of the team member who submits the homework to eCLASS and ID is his or her ID number.

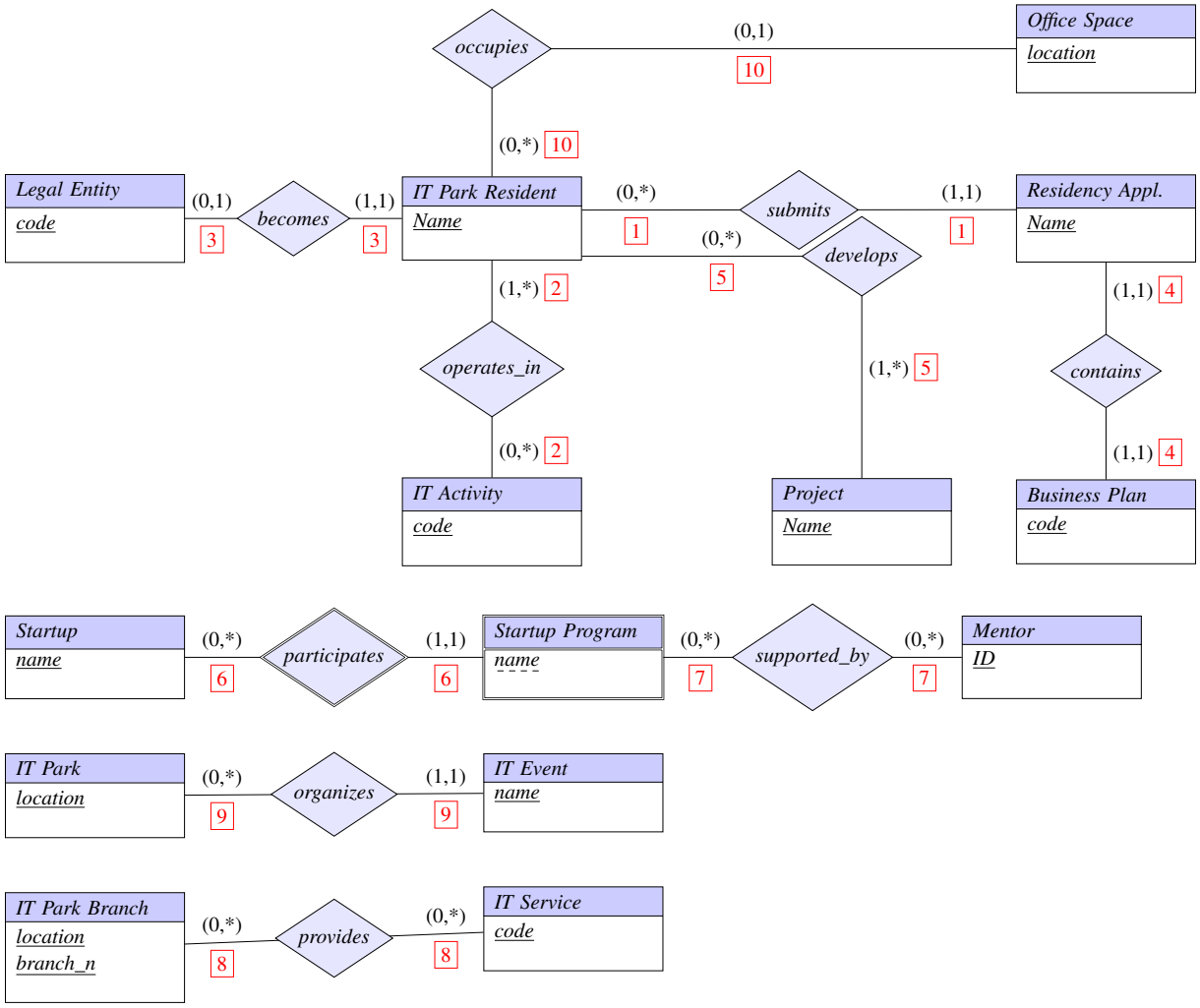
5. PLAGIARISM WARNING! The homework must be your own original work. Homework that containts elements of **plagiarism or cheating**, either among teams or from existing material (e.g., from the Internet, etc.) implies a mark 0/100 **for all teammates**. (Equivalently, **all team members will not be allowed to enter the next in-class quiz**.) You may not copy source code (LaTeX, diagrams, etc.) or any other material from other people or the web. Moreover, dishonesty while discussing an academic integrity issue (i.e. lying to course staff) can result in a 'F' in the course. All violations of this collaboration policy will be referred to the appropriate University disciplinary board, with possible additional disciplinary action. Please ask the instructor (not TA!) if you are unsure.

5.1 You *are* allowed to use generative AI tools provided that you:
⇒ 1. declare type and model (examples: ChatGPT; GPT-5.6 Luna; Gemini 1.5 Pro; Claude 3.5 Sonnet; etc.) at the beginning of each exercise or subtask where you use it, and
⇒ 2. write the prompt and the complete promt's history, if successive steps of refinements are used.

[1] Among others are Texlive, Mactex, and Miktex.

PROBLEMS AND EXERCISES:

Exercise 1: Consider the following E-R diagram (primary keys underlined; ignore numbers in square boxes):



Do the following:

1. Draw an optimized logical (relational) schema in 3NF, extended by adding **at least two attributes** to each relation obtained from mapping each entity set in the diagram.

   Requirement: Two attributes shall be multivalue, and at least two attributes shall be composite.

IMPORTANT: Stick to use the notation in the illustrative schema in Figure 1. (You might wish to refer to the equivalent E-R diagram for company DB—see Homework Assignment 1—to better understand how the logical schema in Figure 1 was mapped from it.) If necessary, manually tune your LaTeX source by promt-engineering an AI tool so to obtain the desired notation. In this exercise, you can **not** ignore (min, max) constraints.

Exercise 2: Consider the logical schema you produced in Exercise 1. Do the following:

1. Write the schema in SQL standard (SQL-92).

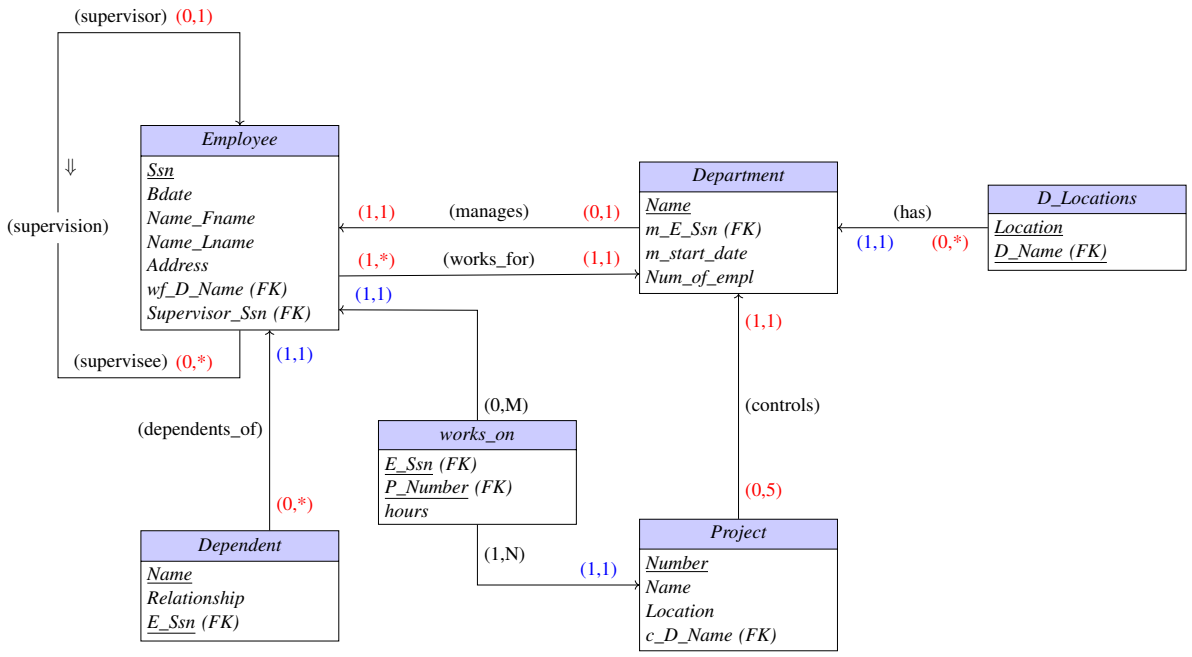


Figure 1: 3NF- Normalized Logical Schema for the company DB (use it as reference for notation).

IMPORTANT: Write each table by using LaTeX's package "listings" set for SQL:

```
\usepackage{listings}
\lstset{language=SQL}
```

List all SQL tables in your document, and **make clear how each table models the** (min, max) **constraints related to it**. Ask your preferred AI tool(s) to generate additional **minimal SQL code** — for instance, triggers — to model the cardinality mapping you don't know how to model by using the in-table standard constraints:

```
UNIQUE Constraint
PRIMARY KEY Constraint
FOREIGN KEY Constraint
NOT NULL Constraint
```

Exercise 3: Consider the SQL tables and related triggers (if any) you produced in Exercise 2.
Do the following:

1. Create a database schema called "DB-HW2" in MySQL.
2. Import all the SQL tables you produced into DB-HW2 schema. To avoid any compilation error, edit the SQL code you wrote in LaTeX to match MySQL's syntax. Debug if necessary.
3. Create a file with name 00X-ID-DDL (00X is the section of the team member who submits the homework to eCLASS; ID is his or her ID number) and extension .sql. This file shall contain all the SQL tables you effectively imported into MySQL. This file shall be included in the zipped archive to submit.

4. Insert, for each SQL table you effectively imported into MYSQL, **ten rows of data** ("data instance"). You might wish to use generative AI tools to speed up this task.

5. Create a file with name 00X-ID-DML (00X is the section of the team member who submits the homework to eCLASS; ID is his or her ID number) and extension .sql. This file shall contain all the statements of type:

   ```
   INSERT INTO ... VALUES ();
   ```

   you used to populate the database with data for all the SQL tables you effectively imported into MYSQL. This file shall be included in the zipped archive to submit.

IMPORTANT: The files 00X-ID-DDL.sql and 00X-ID-DML.sql shall be neat and **contain code commented by you – not by AI!** Moreover, the files shall **not contain auto-generated system code – generated by** MYSQL**!** In other words, if you create these file by exporting from MYSQL the SQL tables (DDL data) and data instance (DML data), you need to "polish" them and clean out all code not striclty related to the logical schema produced in Exercise 2, eventually extended with triggers.

FINAL REMARK: Be ready to complete any part of the tasks described above during an in-class quiz.

End of Homework Assignment 2